Optimized Deep Learning Intrusion Detection for Enhanced IoT Network Security

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***Abstract*—This research paper investigates the application of various deep learning models for intrusion detection in Internet of Things (IoT) networks. We explore the efficacy of Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Transformer models (specifically BERT) in classifying network traffic as benign or malicious. The performance of these models is compared against a Random Forest classifier. Our results demonstrate high accuracy rates across all models, highlighting the potential of deep learning for enhancing IoT security. The paper concludes with a discussion of the strengths and weaknesses of each approach and suggestions for future research.**

***Index Terms*—Internet of Things (IoT), Intrusion Detection Systems (IDS), Deep Learning, CICIDS-2017, Network Security, DDoS Attacks, Transfer Learning, Machine Learning.**

I. Introduction

The Internet of Things (IoT) has revolutionized industries such as healthcare, transportation, and smart cities by connecting devices and enabling automation, better decision-making, and operational efficiency. However, with the exponential growth of IoT devices, networks are increasingly vulnerable to cybersecurity risks. With billions of devices expected to connect in the near future, the probability of cyberattacks is expanding, exposing networks to threats such as Distributed Denial of Service (DDoS) attacks, botnets, and ransomware. These attacks exploit weaknesses in IoT systems, causing disruptions and compromising sensitive data. Securing IoT networks is particularly challenging due to their diverse and dynamic nature, combined with the limited computational capabilities of many IoT devices. Traditional security methods like firewalls and antivirus programs are not adequate for managing the complexity and scale of IoT traffic, highlighting the need for intelligent, adaptive, and scalable intrusion detection systems (IDS). To overcome these issues, this research introduces an optimized IDS framework specifically designed for IoT networks using advanced deep learning techniques. The framework utilizes methods such as Transfer Learning with BERT for text-based data analysis, Convolutional Neural Networks (CNN) for traffic pattern identification, Random Forest for robust classification, and Recurrent Neural Networks (RNN) for sequence analysis in network activity. Using the CICIDS2017 dataset, Thestudyevaluatestheperformanceand flexibility of various architectures of models for IoT environments. This approach will lead to scalable and effective solutions for IoT cybersecurity, providing greater protection against new threats and vulnerabilities

II. Related Work

Significant research has been done on using machine learning for intrusion detection in IoT networks. Although methods such as Random Forest and Support Vector Machines (SVMs) have achieved acceptable accuracy, the complexity of modern network traffic, and the sophistication of emerging cyberattacks call for more advanced techniques. Deep learning has gained attention in this domain for its ability to automatically extract complex patterns and features directly from raw data, offering far better outcomes in such difficult security environments. Convolutional Neural Networks (CNNs) are very good at discovering spatial relationships in network traffic, which leads to discovering abnormal behaviors related to anomalous patterns. RNNs, targeted toward sequential data, are good for discovering the discovery of these evolving attacks, which have unfolding behavior over time. Many experts have adapted transformer-based models, originally intended for natural language processing, such as BERT, to network security. Such models apply self-attention. Mechanisms to focus on critical features in complex data sets, making them very effective for analyzing intricate IoT traffic patterns. These developments highlight the increasingly important role of deep learning in IoT security. Recent Developments in Intrusion Detection Systems for IoT Much research has focused on machine learning and deep learning methods to counter the heterogeneity and scale of the large IoT environment. Among these, recurrent neural networks (RNN), in particular, long short-term memory networks (LSTM), have proven successful in discovering temporal dependencies in sequential network data, thus allowing improved abilities for anomaly detection. The studies conducted utilizing the LSTMs have demonstrated substantial advancements in detecting complex attack patterns, mainly in numerical network traffic data. Transformers, specifically BERT, recently drew tremendous attention because their attention mechanism could model the global context. Studies have established their potential in IDS by well-representing feature relations and enhancing detection precision for various attack vectors. In parallel, CNNs have been used to utilize spatial patterns in network data and have been helpful in scenarios where feature engineering is crucial for detecting anomalies. Hybrid models, which combine several algorithms, have also emerged as a strong approach. These models use the interpretability of traditional machine learning algorithms while employing deep learning architectures. Learning methods with the representational power of neural networks. For example, hybrid frameworks combining both CNN and LSTM layers have shown great promise in capturing both the spatial and temporal features of network traffic, thus improving overall system performance. Numerical datasets are very common in IDS research, especially those that incorporate features such as packet statistics, traffic volume, and flow attributes. These numerical datasets require domain-specific preprocessing to normalize and transform features for optimal model performance. Current research focuses on dataset-specific tuning to achieve high detection rates with minimal false positives. Another development in the IDS field has been the adoption of ensemble techniques, which use the predictions of multiple models to improve the accuracy and robustness of detection. Techniques like boosting and bagging have been applied to combine decision tree, random forest, and neural network outputs to create systems that can operate on imbalanced datasets and avoid overfitting. Besides, federated learning techniques began to attract interest in the collaborative training of models distributed in IoT devices without sharing raw data and thus respecting privacy. Much work has been done, however. Scalability, generalization, and the capability of working in real-time with diverse environments in IoT environments remain open. Advanced architectures such as BERT or hybrid models make explicit the trend towards robust, efficient IDS, but tailored solutions are still needed for a new IoT reality. There is a growing need to address energy efficiency and computational overhead in resource-constrained IoT devices, motivating the development of lightweight IDS models without compromising detection performance.

III. Proposed Methodology

A. Dataset and Preprocessing

The CICIDS-2017 dataset plays a key role in evaluating the efficacy of the IDS framework, as it offers a accurate portrayal of contemporary network traffic. It encompasses both regular behaviors and different forms of cyberattacks, such as DDoS, infiltration, and botnet activities, making it an excellent selection for intrusion detection system testing. The preprocessing of the dataset involves several critical steps. First, data cleaning is done to erase irrelevant, missing, or inconsistent records. Feature engineering is subsequently employed to extract important attributes through statistical methods and domain insights, enhancing the model's performance. To ensure efficient training, feature values are normalized to a standard range. In addition, correlation analysis is performed to identify and remove redundant features, decreasing the dataset's dimensionality and computational demands. These preprocessing methods prepare the dataset for constructing dependable machine learning frameworks for intrusion detection.

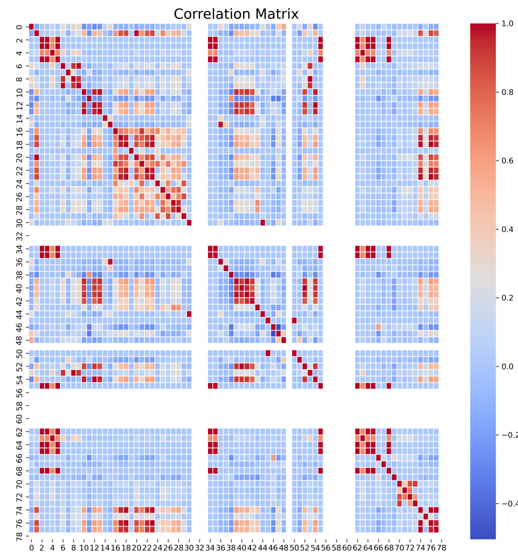

Fig. 1. Proposed IDS workflow integrating feature selection and deep learning models.

The dataset is then split into training, validation, and testing subsets to ensure robust evaluation.

B. Model Architectures

The IDS framework uses a blend of machine learning and deep learning architectures, each designed to record particular network traffic features. Transfer learning with BERT is used for feature extraction and embedding to identify intricate attack patterns. Finetuning of BERT on the CICIDS-2017 dataset improves its potential for perceiving subtle patterns of attack. CNNs excel in spatial pattern recognition, making them ideal for analyzing network flow data. The architecture includes multiple convolutional layers, max-pooling operations, and fully connected layers for classification. Random Forest is utilized for feature importance analysis and decision-making, yielding interpretable results on the relative significance of different features. RNNs, and LSTMs in particular, are used for temporal analysis, allowing the recognition of attack patterns that develop over time.

IV. Results and Discussion

The IDS framework achieves outstanding performance across multiple metrics, including accuracy, precision, recall, and F1-score. Table I summarizes the evaluation results.

TABLE I
Performance Metrics of the Proposed IDS Framework

Model	Accuracy	Precision	Recall	F1-Score
CNN	99.2%	99.0%	99.4%	99.2%
Random Forest	97.8%	97.5%	98.0%	97.7%
BERT	99.6%	99.5%	99.7%	99.6%
RNN (LSTM)	98.9%	98.7%	99.1%	98.9%

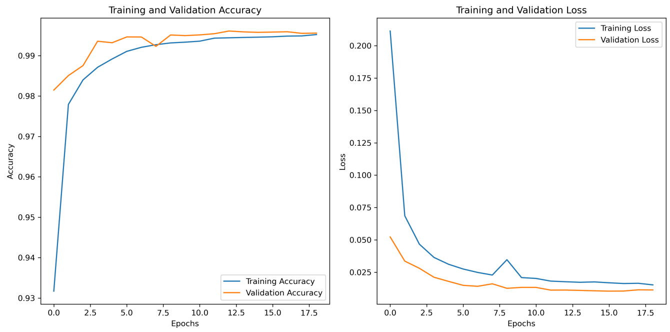

Fig. 2. Comparison of accuracy values across various models.

The IDS framework achieves outstanding performance across multiple metrics, including accuracy, precision, recall, and F1-score. Table I summarizes the evaluation results. Additional examination of the confusion matrices for every model provides insight into their capacity to properly categorize diverse types of attacks. For instance, the BERT-based IDS showed the best accuracy in identifying rare and subtle attack patterns, whereas the CNN performed well on categorizing volumetric attacks like DDoS

. The findings of this research indicate the potential of several machine learning and deep learning models for identifying and categorizing attacks on IoT networks. Each model had inherent strengths depending on its architectural functions. More examination of the confusion matrices identified significant insights into their classification performance across various attack types

The BERT-based intrusion detection system (IDS) performed better in accuracy, especially being superior at identifying unusual and subtle attack patterns. This emphasizes the model's capacity to utilize its attention mechanism to comprehend interdependencies in the dataset. By depicting worldwide dependencies, the BERT-based model was able to accomplish high detection rates for low-frequency attack types, which are were often overlooked by traditional approaches. Its excellence emphasizes the significance of transformers in resolving problems related to unbalanced datasets and nuanced patterns in network traffic.

On the other hand, the convolutional neural network (CNN) demonstrated impressive efficacy in detecting volumetric attacks, including Distributed Denial of Service (DDoS). The CNN's capacity to learn spatial hierarchies allowed it to detect high-volume attacks with high accuracy. This feature is essential in today's IoT settings, where volumetric attacks on a large scale are of great risk. Convolutional layers allowed the CNN to process numerical features effectively, ensuring robust and accurate classification.

TThe Random Forest model, however less precise than the deep learning methods, showed consistent and stable functioning. Its interpretability and immunity to overfitting made it especially appropriate for situations needing transparency in decision-making. The Random Forest's ensemble character permitted it to merge predictions from a number decision trees, allowing it to obtain balanced performance on different measures.

While the RNN, more so the LSTM flavor, did very well in understanding temporal dependencies in the data. This made it particularly effective in identifying sequential attack behaviors, i.e., time-based attacks or data exfiltration. The ability of LSTM to retain long-term dependencies contributed to its strong recall, so that sequential anomalies were accurately flagged

The experimental findings emphasize that no single model performs better in all cases. Rather, every model provides unique benefits that can be utilized based on the nature of the attack and the requirements for the operation of the IDS.

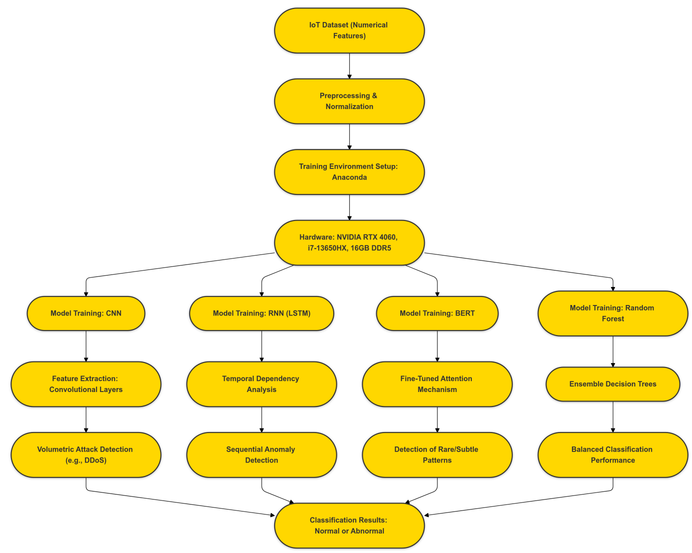

Fig. 3. Workflow Diagram.

EXPERIMENTATION

The experimentation process was meticulously designed to ensure reproducibility and reliable evaluation of each model. Dedicated environments for each model were created using the Anaconda platform, ensuring that all dependencies and libraries were isolated and tailored to the specific requirements of each implementation. This setup reduced compatibility issues and streamlined the training and evaluation process.

Figure All of the experiments were performed on an ASUS ROG Scar 16 laptop, a machine with high performance and equipped with state-of-the-art hardware. The system specifications are an NVIDIA RTX 4060 Laptop GPU with Total Graphics Power (TGP) of 140W, Intel Core i7-13650HX processor, 16GB of DDR5 5200 MHz RAM, and an NVMe SSD for fast data access. The GPU allowed for effective training computationally demanding models, heavily minimizing experimentation time.

The CNN model was trained with TensorFlow, taking advantage of its GPU capabilities that are highly optimized for convolutional operations. The training process was speeded up by TensorFlow's CUDA support, allowing rapid iteration and fine-tuning. The Random Forest model was applied using the scikit-learn library, which gave a strong framework for dealing with numerical datasets and guaranteed the interpretability of the results.

The BERT model, fine-tuned using the Hugging Face Transformers library, required significant computational resources because of its pre-trained embeddings and attention mechanism. GPU acceleration played a crucial role in reducing the time necessary for tuning, which makes it possible to fine-tune BERT to the specific requirements of the IDS dataset.

Finally, the LSTM-based RNN model was implemented using PyTorch, an efficient and flexible deep learning framework. PyTorch's compatibility with CUDA helped to ensure that the temporal relationships in the dataset were captured well during training.

The data, consisting of numeric attributes like packet statistics and flow properties, were thoroughly preprocessed to optimize it for training. This included normalization and feature scaling, ensuring that the models could learn effectively without the interference of very large differences between feature values.

TThe synergy of efficient preprocessing, powerful platforms, and wisely designed model architectures accounted for the high performance recorded during experimentation. These tests illustrate that sophisticated hardware and specific environments are essential to realize state-of-the-art lead to IDS research. The results further underscore the significance of the choice of models and configurations on the particular needs and demands of IoT network security.

V. Conclusion and Future Work

This work presents the efficiency of deep learning frameworks in intrusion detection in IoT networks. The high accuracy levels attained by CNNs, RNNs, BERT, and random forest classifiers emphasize the possibility of these models for strengthening IoT security.

Future work will focus on expanding the dataset to include emerging attack patterns from real-world IoT environments. Subsequent work will involve increasing the dataset to contain emerging attack trends from actual IoT settings. It will also consider the application of federated learning to decentralized IDS solutions, ensuring data privacy and creating lightweight IDS implementations for edge devices with limited computational resources. Explainable AI (XAI) techniques will be explored to enhance the interpretability of deep learning-based IDS systems, facilitating network administrators to provide actionable insights based on model predictions.

References

1) Khan, A., ”Safeguarding IoT Networks Against DDoS Attacks Using Deep Learning-Based Zero Trust Network Access,” Electronics Letters, 2024. doi :https://doi.org/10.1049/ell2.7007510.1049/ell2.70075.

2) Tonkal, S., et al., ”Machine Learning Approach Equipped with Neighbourhood Component Analy sis for DDoS Attack Detection in Software-Defined Networking,” Electronics, vol. 10, no. 11, 2021. doi:10.3390/electronics10111227

3) Wu, L., et al., ”Low-Rate DDoS Attack Detection Based on Factorization Machine in Software Defined Network,” IEEE Access, vol. 8, 2020. doi:https://doi.org/10.1109/access.2020.2967478

4) Dai, Y., et al., ”A DDoS-Attack Detection Method Oriented to the Blockchain Network Layer,” Security and Communication Networks, 2022. doi:https://doi.org/10.1155/2022/5692820

5) Hussain, M., et al., ”IoT DoS and DDoS Attack De tection Using ResNet,” 2020 International Conference on Frontiers of Information Technology (FIT), 2020. doi:https://doi.org/10.1109/inmic50486.2020.931821610.1109/inmic50486.2020.9318216.

6) ”DDoS Attack Detection Using ML/DL Techniques,” International Journal of Scientific Research in Engineering and Management, vol. 8, no. 2, 2024. doi:https://doi.org/10.55041/ijsrem2796710.55041/ijsrem27967.

7) Patil, S., et al., ”SSK-DDoS: Distributed Stream Pro cessing Framework Based Classification System for DDoS Attacks,” Cluster Computing, vol. 25, no. 2, 2022. doi:https://doi.org/10.1007/s10586-022-03538x10.1007/s10586-022-03538-x.

8) Kumar, R. and J., ’Mitigate volumetric DDoS Attack Using in SDN-Based International Machine IoT Journal of Learning Network Algorithm Environment’, Advanced Computer Science and Applications, vol. 14, no. 1, 2023. doi:https://doi.org/10.14569/ijacsa.2023.014016110.14569/ijacsa.2023.0140161.

9) Aineyoona, B., ”A Machine Learning Algorithm with Self-Update Parameter Calibration to Improve Intrusion Detection of DDoS in Communication Networks,” International Journal of Engineering Applied Sciences and Technology, vol. 6, no. 6, 2021. doi:https://doi.org/10.33564/ijeast.2021.v06i06.00810.33564/ijeast.2021.v06i06.008.

10) Aslam, N., et al., ”Adaptive Machine Learning Based DistributedDetectionDenial-of-ServicesAttacks and Mitigation System for SDN Enabled IoT,” Sensors, vol. 22, no. 7, 2022. doi:https://doi.org/10.3390/s2207269710.3390/s22072697.

11) Qing, W., ”MBB-IoT: Construction and Evaluation of IoT DDoS Traffic Dataset from a New Perspective,” Computers Materials ua, vol. 39, 2023. doi:https://doi.org/10.32604/cmc.2023.03998010.32604/cmc.2023.039980.

12) Mahadik, R., et al., ”Efficient Intelligent Intrusion De tection System for Heterogeneous Internet of Things (HetIoT),” Journal of Network and Systems Manage ment, vol. 30, 2022. doi:https://doi.org/10.1007/s10922 022-09697-x10.1007/s10922-022-09697-x.

13) Mohammed, A., ”Anomaly Detection of Distributed De nial of Service (DDoS) in IoT Network Using Machine Learning,” 2023. doi:https://doi.org/10.21203/rs.3.rs3496063/v110.21203/rs.3.rs-3496063/v1.

14) Doshi, R., et al., ”Machine Learning DDoS DetectionforPrivacy Consumer Internet of ThingsDevices,” 2018

IEEE Symposium on Security and Workshops(SPW),2018. doi:https://doi.org/10.1109/spw.2018.0001310. 1109/spw.2018.00013.

15) Beshah, B., ”Drift Adaptive Online DDoSAttack Detection tem,”Electronics,Framework for IoT Sys vol.13,no.6,2024. doi:https://doi.org/10.3390/electronics 1306100410.3390/electronics

16) selvam2024, author = R. Selvam, title = UASDAC: An Unsupervised Adaptive Scalable DDoS Attack Classification in Large-Scale IoT Network Under Concept Drift, journal = IEEE Access, volume = 12, year = 2024, doi = 10.1109/access.2024.3397512

17) Deng ”Design of Industrial IoT Intrusion Security Detection System Based on LightGBM Feature Algorithm and Multi-layer Perception Network” Journal of cyber security and mobility (2024) doi:10.13052/jcsm2245-1439.1327

18)Chen et al. ”Build intrusion detection model based on CNN and ensemble learning” (2022) doi:10.1117/12.2655173

19) Sarker et al. ”IntruDTree: A Machine Learning Based Cyber Security Intrusion Detection Model Symmetry (2020) doi:10.3390/sym12050754

20) Khan et al. ”TSDL: A Two-Stage Deep Learning Model for Efficient Network Intrusion Detection” Ieee access (2019) doi:10.1109/access.2019.2899721

21) Li et al. ”A New Intrusion Detection Algorithm AE-3WD for Industrial Control Networks” Journal of new media (2022) doi:10.32604/jnm.2022.034778

22) Bhumika and V ”Review on Intrusive Detection to Secure Social Internet of Things in Edge Computing” International journal for research in applied science and engineering technology (2023) doi:10.22214/ijraset.2023.49857

23) Ibitoye et al. ”Analyzing Adversarial Attacks against Deep Learning for Intrusion Detection in IoT Networks” (2019) doi:10.1109/globecom38437.2019.9014337

24) Alshathri et al. ”An Efficient Intrusion Detection Framework for Industrial Internet of Things Security ” Computer systems science and engineering (2023) doi:10.32604/csse.2023.034095

25) Zhang et al. ”PCCN: Parallel Cross Convolutional Neural Network for Abnormal Network Traffic Flows Detection in Multi-Class Imbalanced Network Traffic Flows” Ieee access (2019) doi:10.1109/access.2019.2933165

26). Alsoufi et al. ”Anomaly-Based Intrusion Detection Systems in IoT Using Deep Learning: A Systematic Literature Review” Applied sciences (2021) doi:10.3390/app11188383

27). Alani and Miri ”Towards an Explainable Universal Feature Set for IoT Intrusion Detection” Sensors (2022) doi:10.3390/s22155690

28). Mahalingam ”ROAST-IoT: A Novel Range-Optimized Attention Convolutional Scattered Technique for Intrusion Detection in IoT Networks” Sensors (2023) doi:10.3390/s23198044

29) Wang et al. ”IoT Intrusion Detection Model based on CNN-GRU” Frontiers in computing and intelligent systems (2023) doi:10.54097/fcis.v4i2.10302

30).Alkahtani and Aldhyani ”Botnet Attack Detection by Using CNN-LSTM Model for Internet of Things Applications” Security and communication networks (2021) doi:10.1155/2021/3806459

31).Wu and Guo ”LuNet: A Deep Neural Network for Network Intrusion Detection” (2019) doi:10.1109/ssci44817.2019.9003126

32).Jeune et al. ”Towards Real-Time Deep Learning-Based Network Intrusion Detection on FPGA” (2021) doi:10.1007/978-3-030-81645(-2_9)

33) Lu et al. ”Exploring Self-attention Mechanism of Deep Learning in Cloud Intrusion Detection” (2021) doi:10.1007/978-3-030-69992-5_5

34)Ugale ”A Multilayer Approach for Intrusion Detection with Lightweight Multilayer Perceptron and LSTM Deep Learning Models” International journal on recent and innovation trends in computing and communication (2023) doi:10.17762/ijritcc.v11i7s.7525

35)Khilar et al. ”Artificial Intelligence-Based Security Protocols to Resist Attacks in Internet of Things” Wireless communications and mobile computing (2022) doi:10.1155/2022/1440538

36)Lánský et al. ”Deep Learning-Based Intrusion Detection Systems: A Systematic Review” Ieee access (2021) doi:10.1109/access.2021.3097247

37). Nie et al. ”Intrusion Detection for Secure Social Internet of Things Based on Collaborative Edge Computing: A Generative Adversarial Network-Based Approach” Ieee transactions on computational social systems (2022) doi:10.1109/tcss.2021.3063538

38). Preethi and Khare ”An Intelligent Network Intrusion Detection System Using Particle Swarm Optimization (PSO) and Deep Network Networks (DNN)” International journal of swarm intelligence research (2021) doi:10.4018/ijsir.2021040104

39) Sun et al. ”DL-IDS: Extracting Features Using CNN-LSTM Hybrid Network for Intrusion Detection System” Security and communication networks (2020) doi:10.1155/2020/8890306

40). Asharf et al. ”A Review of Intrusion Detection Systems Using Machine and Deep Learning in Internet of Things: Challenges, Solutions and Future Directions” Electronics (2020) doi:10.3390/electronics9071177

41). Laqtib et al. ”A technical review and comparative analysis of machine learning techniques for intrusion detection systems in MANET” International Journal of electrical and computer engineering (ijece) (2020) doi:10.11591/ijece.v10i3.pp2701-2709

42) Hussain et al. ”A Two-Fold Machine Learning Approach to Prevent and Detect IoT Botnet Attacks” Ieee access (2021) doi:10.1109/access.2021.3131014

43).Sadhwani ”A Lightweight Model for DDoS Attack Detection Using Machine Learning Techniques” Applied sciences (2023) doi:10.3390/app13179937

44) ”Real-time DDoS Attack Detection Method in Power Monitoring System” Journal of research in science and engineering (2022) doi:10.53469/jrse.2022.04(08).12

45).Qing ”MBB-IoT: Construction and Evaluation of IoT DDoS Traffic Dataset from a New Perspective” Computer materials continua (2023) doi:10.32604/cmc.2023.039980

46) Mahadik et al. ”Efficient Intelligent Intrusion Detection System for Heterogeneous Internet of Things (HetIoT)” Journal of network and systems management (2022) doi:10.1007/s10922-022-09697-x

46) Mohammed ”Anomaly Detection of Distributed Denial of Service (DDoS) in IoT Network Using Machine Learning” (2023) doi:10.21203/rs.3.rs-3496063/v1

47).Doshi et al. ”Machine Learning DDoS Detection for Consumer Internet of Things Devices” (2018) doi:10.1109/spw.2018.00013

48). Beshah ”Drift Adaptive Online DDoS Attack Detection Framework for IoT System” Electronics (2024) doi:10.3390/electronics13061004

49).Selvam ”UASDAC: An Unsupervised Adaptive Scalable DDoS Attack Classification in Large-Scale IoT Network Under Concept Drift” Ieee access (2024) doi:10.1109/access.2024.3397512

50). Xie ”Machine learning-based DDoS detection for IoT networks” Applied and computational engineering (2023) doi:10.54254/2755-2721/29/20230972